

Bowen Yuan

PHD CANDIDATE · HEAD TEACHING ASSISTANT

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SKILLS

- **Programming & Data:** Python, Java, SQL, Bash, Numpy, Pandas
- **Machine Learning & Multimodal AI:** PyTorch, Deep Learning, Transformer Models, Multimodal Learning, Computer Vision, Generative AI, Knowledge Distillation
- **LLMs, Agentic AI:** LLM/LVLM Fine-Tuning, LLM Post-Training, Reinforcement Fine-Tuning, Tool Use, Prompt Engineering, Grounded Generation, Skill-based Agent Design
- **Evaluation & Data-Centric ML:** Model Evaluation, Benchmarking, Error Analysis, Ablation Studies, Offline Experimentation, Active Learning, Coreset Selection, Data Distillation, Domain Adaptation, Test-Time, Adaptation
- **Tools:** Git, Linux, Jupyter, Weights & Biases, LaTeX

EXPERIENCE

Data Science Team, The University of Queensland

Jun. 2023 – Present

Research Assistant

Brisbane, Australia

- Led research on **machine learning**, **computer vision**, and **agricultural AI** for robust, efficient real-world systems.
- Developed a **self-training vegetation segmentation pipeline** with pseudo-labeling, lightweight transformer design, and a **dashboard** for data and model analysis.
- Conducted **cross-domain**, **ablation**, and **efficiency** experiments to evaluate model robustness, adaptation, and computational trade-offs.

School of EECS, The University of Queensland

Aug. 2021 – Present

HEAD Teaching Assistant

Brisbane, Australia

- Delivered **tutoring**, **marking**, **invigilation**, and **assignment design** for courses including **Database Systems**, **Algorithms & Data Structures**, **High Dimensional Data**, and **Data Science Capstone Project and Thesis**.
- Guided students in applying **machine learning**, **data analysis**, and **system design** to technical coursework and project-based problem solving.
- Supported course delivery through **technical mentoring**, **assessment coordination**, and **learning material design** for diverse student cohorts.

EDUCATION

The University of Queensland

Jan. 2024 – Present

DOCTOR OF PHILOSOPHY (PHD) IN COMPUTER SCIENCE

Brisbane, Australia

- Research in machine learning, dataset distillation, large vision language models, supervised by Prof. Helen Huang, and Dr. Zijian Wang.

The University of Queensland

Feb. 2020 – Nov. 2023

BACHELOR OF COMPUTER SCIENCE (HONOURS CLASS I)

Brisbane, Australia

- GPA 6.7/7.0; Dean's Commendation for Academic Excellence; thesis: Automatic plot segmentation and position from remote sensing images

SELECTED PROJECTS

Agricultural AI Dashboard Platform

Research Project

- Built an integrated **agricultural AI dashboard** for end-to-end **wheat yield prediction**, **disease detection**, and **cross-domain segmentation**, supporting data analysis, model development, and result visualization in a unified workflow.
- Designed the framework to include **yield prediction**, **vision-based disease detection**, and a **self-training segmentation pipeline** with training, inference, and evaluation modules, together with interactive dashboard components for comparing models and experiments.
- Demonstrated effectiveness through segmentation experiments, achieving **improved cross-domain performance** and **reduced computational cost** over baseline methods in the wheat segmentation study.

Agricultural VLMs & Agents

Research Project

- Developed **LLM-based agricultural agents** for diverse downstream tasks, including **disease identification**, **quantification**, **diagnosis**, and **recommendation generation** in agricultural decision-support settings.
- Collected multimodal datasets with over 60,000 samples and built a **post-training framework** that couples **Supervised Fine-Tuning (SFT)** with **RL-based GRPO** for training LVLMs to provide wheat diagnosis.
- Designed an agent workflow with **task orchestration**, **tool use**, and **multi-task problem solving**, together with curated reward functions for multiple agricultural tasks.

RL Post-Training for LVLMs

Research Project

- Developed **LASER**, a **reinforcement post-training method** that mitigates visual forgetting by regulating visual attention throughout long-horizon multimodal reasoning.
- Post-trained Qwen2.5-VL-7B-Instruct using cold-start training followed by GRPO on 45,000 multimodal reasoning samples.
- Designed a Visual Grounding Reward to preserve task-relevant visual attention and a Sink Suppression Reward to penalize attention collapse.

SELECTED PUBLICATIONS

- **Bowen Yuan**, Zijian Wang, Mahsa Baktashmotlagh, Yadan Luo, and Zi Huang. *Color-oriented redundancy reduction in dataset distillation*. Advances in Neural Information Processing Systems (**NeurIPS 2024**). [\[PDF\]](#)
- **Bowen Yuan***, Selena Song*, Javier Fernandez, Yadan Luo, Mahsa Baktashmotlagh, and Zijian Wang. *WisWheat: A Three-Tiered Vision-Language Dataset for Wheat Management*. Proceedings of the 33rd ACM International Conference on Multimedia (**ACM MM 2025**). [\[PDF\]](#)
- **Bowen Yuan***, Yuxia Fu*, Zijian Wang, Yadan Luo, and Zi Huang. *SCORE: Soft Label Compression-Centric Dataset Condensation via Coding Rate Optimization*. IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV 2026**). [\[PDF\]](#)
- **Bowen Yuan**, Zijian Wang, Yadan Luo, Shijie Wang, and Zi Huang. *LASER: A Corrective Lens for LVLMs via Visual Attention Preservation and Sink Suppression*. The 19th European Conference on Computer Vision (**ECCV 2026**). [\[PDF\]](#)